Course: BM20A4703 Partial Differential Equations with Applications.
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Worksheet 2

Exercise 2.1

Solve the following PDEs using the method of characteristics:

(1) [10] For $x \in \mathbb{R}, t > 0$,
$$2\frac{\partial u}{\partial x} + \frac{\partial u}{\partial t} = 0, \quad u(x,0) = \frac{1}{1+x^2}.$$

(2) [10] For $x, y \in \mathbb{R}$,
$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = xe^{-u}, \quad u(x,x^2) = 0.$$

(3) [10] For $x, y \in \mathbb{R}$,
$$u\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 1, \quad u(x,x) = \frac{x}{2}.$$

Exercise 2.2

(1) [10] Prove that if $E(t)$ satisfies $E'(t) \le G(t) + E(t)$ for $t > 0$ and some non-negative G, and $E(0) = 0$ then
$$E(t) \le e^t \int_0^t G(s)\mathrm{d}s \quad \text{for all } t \ge 0.$$

(2) [20] Let $a > 0, R > 0$. Consider the initial-boundary value problem of the transport equation
$$\begin{cases} u_t + au_x = f(x,t), & 0 < x < R, t > 0, \\ u(0,t) = 0, & t > 0, \\ u(x,0) = 0, & 0 < x < R, \end{cases}$$
with a source term f. Use (1) to prove the stability estimate (for $t > 0$)
$$\int_0^R u^2(x,t)\mathrm{d}x \le e^t \int_0^t \int_0^R f^2(x,s)\mathrm{d}x\mathrm{d}s.$$
This inequality implies that the solution depends stably on the source term, in L^2-norm, at least for finite times.
Hint: Using energy method. Multiply by u the equation, integrate over $(0,R)$ and use integration by parts.

Exercise 2.3

[10] Verify by direct calculations that the formula given in the lecture notes $u = g(x - t\mathbf{F}'(u))$ provides an implicit solution for the Cauchy problem
$$u_t + \operatorname{div}\mathbf{F}(u) = 0, \quad u(x,0) = g(x) \quad (x \in \mathbb{R}^n).$$
(Note: $\operatorname{div}\mathbf{F}(u) = \mathbf{F}'(u) \cdot Du$).